Problem:-

Consider the displacement field.

$$\vec{u} = \left[y^2 \hat{i} + 3yz\,\hat{j} + (4+6x^2)\hat{k} \right] \times 10^{-2}$$

What are the rectangular strain components at point $P(1,0,2)$

Soln:-

$$u = y^2 \times 10^{-2}$$
$$v = 3yz \times 10^{-2}$$
$$w = 4 + 6x^2 \times 10^{-2}$$

The displacement gradient is therefore.

$$\nabla\vec{u} = \begin{bmatrix} 0 & 2y & 0 \\ 0 & 3z & 3y \\ 12x & 0 & 0 \end{bmatrix} \times 10^{-2} \quad \& \quad (\nabla\vec{u})^T = \begin{bmatrix} 0 & 0 & 12x \\ 2y & 3z & 0 \\ 0 & 3y & 0 \end{bmatrix} \times 10^{-2}$$

$$\therefore \epsilon = \frac{1}{2}\left[(\nabla\vec{u}) + (\nabla\vec{u})^T \right]$$
$$= \frac{1}{2}\begin{bmatrix} 0 & 2y & 12x \\ 2y & 6z & 3y \\ 12x & 3y & 0 \end{bmatrix} \times 10^{-2}$$

$$\epsilon_{P(1,0,2)} = \frac{1}{2}\begin{bmatrix} 0 & 0 & 12 \\ 0 & 12 & 0 \\ 12 & 0 & 0 \end{bmatrix} \times 10^{-2} = \begin{bmatrix} 0 & 0 & 6 \\ 0 & 6 & 0 \\ 6 & 0 & 0 \end{bmatrix} \times 10^{-2}$$

$\Rightarrow \quad \epsilon_{xx} = 0 \qquad\qquad \epsilon_{xy} = 0$

$\epsilon_{yy} = 6 \times 10^{-2} \qquad\qquad \epsilon_{xz} = 6 \times 10^{-2} \; (\gamma_{xz} = 2\epsilon_{xz} = 12 \times 10^{-2})$

$\epsilon_{zz} = 0 \qquad\qquad \epsilon_{yz} = 0.$

—— x ——

Ques:- The following state of Strain exists at point P.

$$[\epsilon_{ij}] = \begin{bmatrix} 0.02 & -0.04 & 0 \\ -0.04 & 0.06 & -0.02 \\ 0 & -0.02 & 0 \end{bmatrix}$$

In the direction PQ having direction Cosines
$n_x = 0.6$, $n_y = 0$ and $n_z = 0.8$
determine ϵ_{PQ}. Also determine the Cubical dilation at point P.

Soln:- We have been asked to determine ϵ_{PQ} in the direction PQ ie normal Strain in the direction (n_x, n_y, n_z).

The analogy of Stress transformation Can be applied here.

Traction on a plane is $p(\hat{n}) = \tau\,\hat{n}$.
& the Normal Comp. of traction = $p(\hat{n})\cdot\hat{n}$.

ϵ_{pq} = Normal Component of Strain along the $(n_x\, n_y\, n_z)$

$$= (\epsilon\,\hat{n})\cdot\hat{n} = (\epsilon\,\hat{n})^T\,\hat{n}.$$

$$= \left\{ \underset{3\times 3}{\begin{bmatrix} 0.02 & -0.04 & 0 \\ -0.04 & 0.06 & -0.02 \\ 0 & -0.02 & 0 \end{bmatrix}} \underset{3\times 1}{\begin{bmatrix} 0.6 \\ 0 \\ 0.8 \end{bmatrix}} \right\}^T \underset{3\times 1}{\begin{bmatrix} 0.6 \\ 0 \\ 0.8 \end{bmatrix}}$$

$$= \begin{bmatrix} 0.012 & -0.04 & 0 \end{bmatrix} \begin{bmatrix} 0.6 \\ 0 \\ 0.8 \end{bmatrix}$$

$$= 7.2\times 10^{-3}.$$

(ii) The Cubical dilation at point P

$$\epsilon_V = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz} = 0.02 + 0.06 + 0 = 0.08.$$